

Mars Atmosphere and Dust Sample Return

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MISSION STATEMENT, OBJECTIVES, AND CONSTRAINTS

Although thin, the Martian atmosphere circulates 400 Tg of dust annually, with peaks occurring during annual planet-wide dust storms. Returning dust from the Martian atmosphere would constitute the first return of Mars surface materials (and atmosphere), and an important “first” in understanding the chemistry, mineralogy, and atmospheric composition of the red planet. The objective of this mission is to collect samples of Martian dust and atmosphere, and return the pristine samples to Earth. The mission is to be launched in the end 2028/early 2029 Mars opportunity.

Element of Mission Architecture	Can be Traded	Reason
Mission Concept	Yes	Remain open to alternative approaches
Subject	No	Subject is fixed by mission objectives
Payload	Size and design	Mission objective defines sample return payload
Spacecraft Bus	Yes	Different buses available based on size
Launch System	Yes	Open to any capable, single launch system
Trajectory	Yes	Only limitations on launch window
Grounded System	Yes	Different communications systems available
Communications Architecture	Yes	Different communications systems available
Mission Operations	Yes	Can adjust automation level and onboard processing

Table 1. Mission architecture elements.

The sample return of the atmosphere and atmospheric dust is the primary objective of this mission. Due to the difficulty of this unprecedented mission, secondary objectives are not present, though mission constraints do exist. The mission is limited to a single launch vehicle and mission protocol must comply with the Mars Planetary Protection requirements determined by the Planetary Protection category that the sample return mission falls within. As will be detailed in the analysis of the mission options, the mission objectives can be met fully, partially, or not at all. The chosen mission design should be optimized to meet the mission objectives as fully as possible.

MISSION ANALYSIS

Functional Analysis

The Mars sample return mission can be broken down into different levels of a flow diagram to perform functional analysis of the mission.

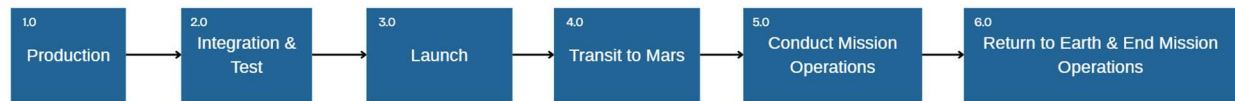


Figure 1. Top level flow.

This top level flow is similar to previous interplanetary missions. A second level flow provides a more detailed breakdown of mission operations. Note the feedback loop on 5.3 Confirm Sample Collection. This is to ensure

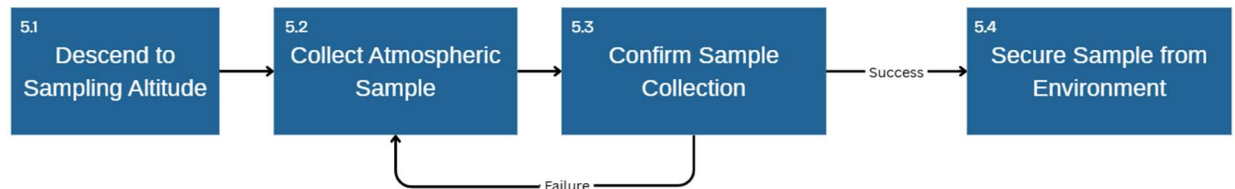


Figure 2. Second level flow.

Especially important for sample return missions is the method of sample collection. The Mars sample return mission is unique due to the presence of an atmosphere, which is the target sampling location set by the mission objectives. The nature of the mission, interplanetary travel for a short sample period, presents difficulty for the sample collection. Either the spacecraft must fully enter and exit Mars' gravity well and spend significant energy in the process or the sample will be taken at extremely high speeds, presenting issues with flows and heating as the spacecraft passes through the Martian atmosphere at such extreme velocities. These sample collection environments present significantly different sample collection methods required. The third level flows below breaks down these sample collection processes further.

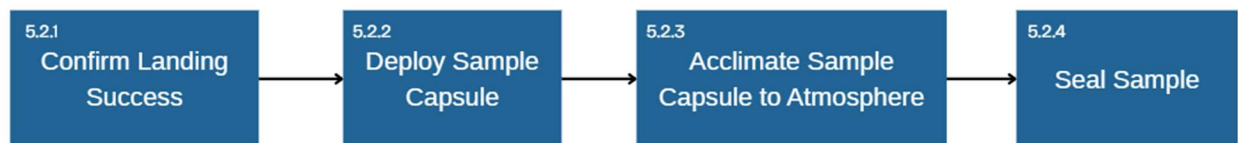


Figure 3. Third level flow for surface sample collection.

For the ground-based collection, the sample capsule needs to acclimate with the environment before the sample is secured. This is similar to the sample collection used by the Perseverance rover currently on Mars.

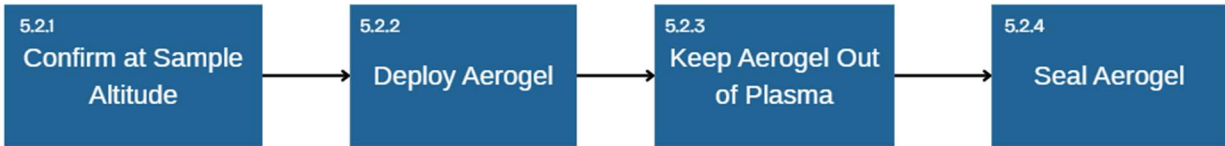


Figure 4. Third level flow for high altitude sample collection.

For a high-altitude sample collection, the sample is also collected at high velocity. The high velocity sample particles can significantly damage and destroy traditional sample capsules, so an aerogel is used instead. The aerogel is an extremely light solid that slows down the particles and absorbs their kinetic energy without damaging the spacecraft or the particles.

Orbits and Trajectories

The window to Mars is already defined by mission constraints and orbital mechanics, but decisions are still left to be made. The trajectory to Mars will define the delta-V budget for the mission, the total mission time, and the trajectory of the spacecraft through the Martian atmosphere. There are three trajectory options for this mission: Hohmann transfer, free return, and Venus gravity assist. The Hohmann transfer is the most commonly used transfer for mission plans to Mars. The Hohmann transfer presents the fastest travel time at the expense of ΔV . The free return trajectory would only be used with high altitude sampling method due to a high velocity atmospheric entry of Mars. However, the free return trajectory also requires the lowest ΔV at Mars. Finally, the Venus gravity assist has the lowest launch energy which allows for larger mass payloads. This comes at the expense of a much longer time of flight. Also, further investigation revealed that the positioning of Earth, Venus, and Mars is not ideal. This is best illustrated by the plots created by Kyle M. Hughes, Peter J. Edelman and James M. Longuski at Purdue University below, where the left plot is for an Earth-Venus-Mars-Earth trajectory and the right plot is for an Earth-Mars-Venus-Earth trajectory.

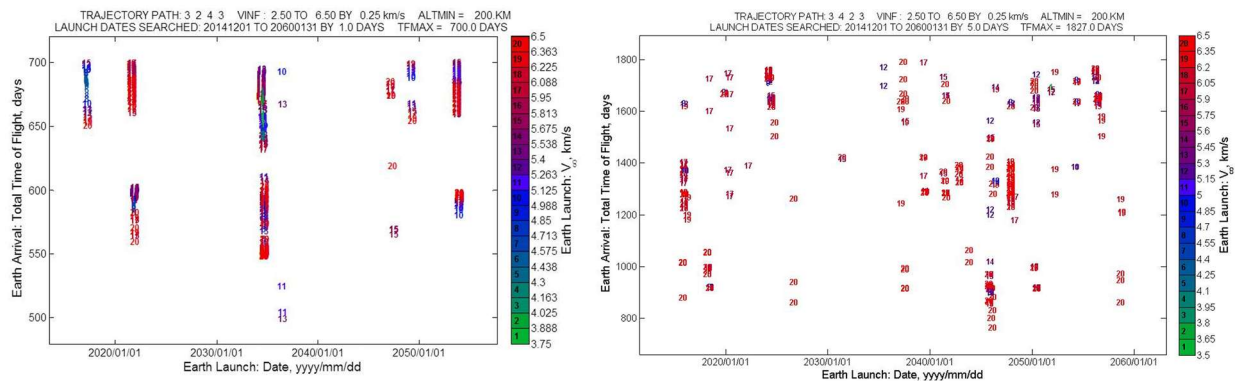


Figure 5. Venus gravity assist total time of flight based on Earth launch date (Hughes, Edelman, Longuski).

Note the distinct lack of viable launch dates in the mission designated launch window of late 2028 and early 2029 and the high velocities required for the closest launches. The launch velocity is denoted with the color and number scale on the right, where a green 1 is lowest and a red 20 is the highest.

Design Decision Matrix

For the design decision matrix, the first decision presented is between the trajectories. Following with the mission constraint of the late 2028 or early 2029 launch window, only the Hohmann transfer and the free return trajectories will be listed. Next are the sampling methods, which are heavily correlated with the mission trajectory. Note the separate lander and lander with rover. The rover option would allow for better sample options if solid samples were included in the mission. The rover could also be left on Mars, leading to more potential scientific benefit from the mission. The communications networks would be correlated with the sampling method. A flyby sampling mission would not require a relay. The launch vehicle is important, especially with the single launch constraint. While many options are becoming available on the market with new developments, this mission will rely on flight tested hardware as much as possible to ensure mission success. The final part of the decision aligns with the final part of the mission, the Earth return. It is important to consider the planetary protection requirements for Mars. While previous sample return missions have dropped samples directly into the Utah desert for collection, alternative, safer return methods must also be explored. This decision matrix will include an ISS rendezvous and a LEO orbit as alternatives to the more traditional sample drop.

Trajectory	Hohmann Transfer	Free Return	-
Sampling Method	Aerogel	Lander-Arm	Lander-Rover
Communications	Direct	Orbital Relay	-
Launch System	SLS	Angara A5	Falcon Heavy
Sample Return	Desert Drop	ISS Rendezvous	LEO Rendezvous

Table 2. Mission design decision matrix.

MISSION DESIGN OPTIONS

Through several design decisions, the design paths have been narrowed to the paths in the table below. This was realized by eliminating nonsensical paths, such as a free return trajectory with a Mars lander. Physically incompatible mission paths, such as the mass heavy rover sampling method and the smaller Angara A5 rocket, were also removed from consideration. For the table and future reference, note the abbreviations of Mars Ascent Vehicle (MAV) and Mars Orbital Vehicle (MOV).

Element	Path 1	Path 2	Path 3
Mission Concept	Mars atmospheric skimming flyby with direct entry return	Mars lander rendezvousing with MOV before direct entry return	Mars rover rendezvousing with MOV with LEO rendezvous return
Sampling Method	Aerogel	Lander arm	Rover
Trajectory	Free return	Hohmann	Hohmann
Launch System	Angara A5	Falcon Heavy	SLS
Communications	Direct	Orbital relay	Orbital relay
Sample Return	Direct entry	Direct entry	LEO rendezvous
Mission Operations	Limited	Limited	Continuous

Table 3. Prioritized mission design paths.

Design path 1 is selected with the philosophy of a simple, faster mission. There are fewer points of failure and omitting the Mars landing results in large mass savings. Design path 2 is designed as a more traditional approach. Design path 3 is chosen with a science first mentality. The mission would leave a rover on Mars after sample collection for further scientific exploration and include a more controlled sample return.

To estimate the spacecraft masses, a parametric method was adopted for the dry mass along with analogous missions. Starting with the payload mass, a survey of similar missions called COMPASS developed by NASA Glenn Center provides a relationship between the payload mass and the total dry mass of the spacecraft. After the dry mass is properly sized, the mission design tool developed previously in this course allows the propellant mass and loaded mass to be determined. For design path 1, a 50 kg payload was chosen based on the Stardust spacecraft. For path 3, the payload was designated as 1 metric ton, based on the previous Mars rover masses. From this process, the propellant mass percentage and dry mass percentage of the spacecraft can be determined, as shown in the table below.

Design Path	Loaded Mass (kg)	Propellant Mass (kg)	Dry Mass (kg)	Propellant Mass (%)	Dry Mass (%)
1	449.9	33.2	416.7	7.4%	92.6%
2	14419.8	9799.8	4620	68.1%	31.9%
3	32785.4	26855.4	5930	81.9%	18.1%

Table 4. Basic mass breakdown of spacecraft.

For *Table 4* above, these are masses are designed to be well within the payload mass that the associated launch system can deliver to Mars. This allows room for mass increase changes later in the design that may be necessary. These numbers only include the spacecraft masses. The masses including the launch systems are 749 metric tons, 1365 metric tons, and 2604 metric tons, for paths 1, 2, and 3 respectively.

Design Path	Payload (%)	Structure (%)	Power (%)	C&DH (%)	Propulsion (%)	ADC (%)
1	18.8%	42.5%	12.5%	5%	11.2%	10%
2	15.2%	45.5%	7.5%	3.7%	25.9%	2.2%
3	25.7%	34.6%	9.8%	3.3%	23.8%	2.8%

Table 5. Sub-system mass percent breakdown of spacecraft dry mass.

The mass estimates for the sub-systems above were calculated using parametric methods, using comparisons to similar missions for each design path. Since each design path was unique, different parametric equations were employed for each design path, using similar missions to that design path.

The trajectory of the mission is especially vital to interplanetary sampling missions due to their effects on return trajectories, mission scheduling, and payload mass upper limits. For design path 1, the free return trajectory is a more complex and time dependent trajectory. This makes the timing more complex but, if executed properly, lowers the requirements of maintaining the system after the Trans-Martian Injection Burn. The architectural result is a single, larger point of failure. The single burn would require 4.76 km/s of ΔV , along with 0.44 km/s of additional ΔV budgeted for course correction maneuvers for a total ΔV of 5.2 km/s.



Figure 6. Free return trajectory depicting Mars contact (left) and Earth return (right).

The free return trajectory is shown above with the leg from Earth to Mars on the left and then the returning leg on the right. The Hohmann transfer is much more common in Mars missions. It offers the benefit of allowing longer stays on Mars, which is greatly beneficial for design paths 2 and 3 that require extended stays on Mars' surface. However, contrasting the free return trajectory, the Hohmann transfer requires two burns for a proper execution. This leaves a point of failure at the start of the return leg for the return burn. The total ΔV for this maneuver is 4.35 km/s for a direct sample return. Including the landing on Mars, the total ΔV increases to 12.25 km/s. If a rendezvous is used instead, the total ΔV increases to 17.75 km/s. It should be noted that technically this is not a Hohmann transfer, which is not possible with the positions of the Sun, Earth, and Mars. Lambert targeting was used to approximate a Hohmann transfer. All the trajectories are simulated with in FreeFlyer for the figures in this report. These trajectories are

low fidelity and designed to illustrate the general trajectories rather than actually map the spacecraft's course.

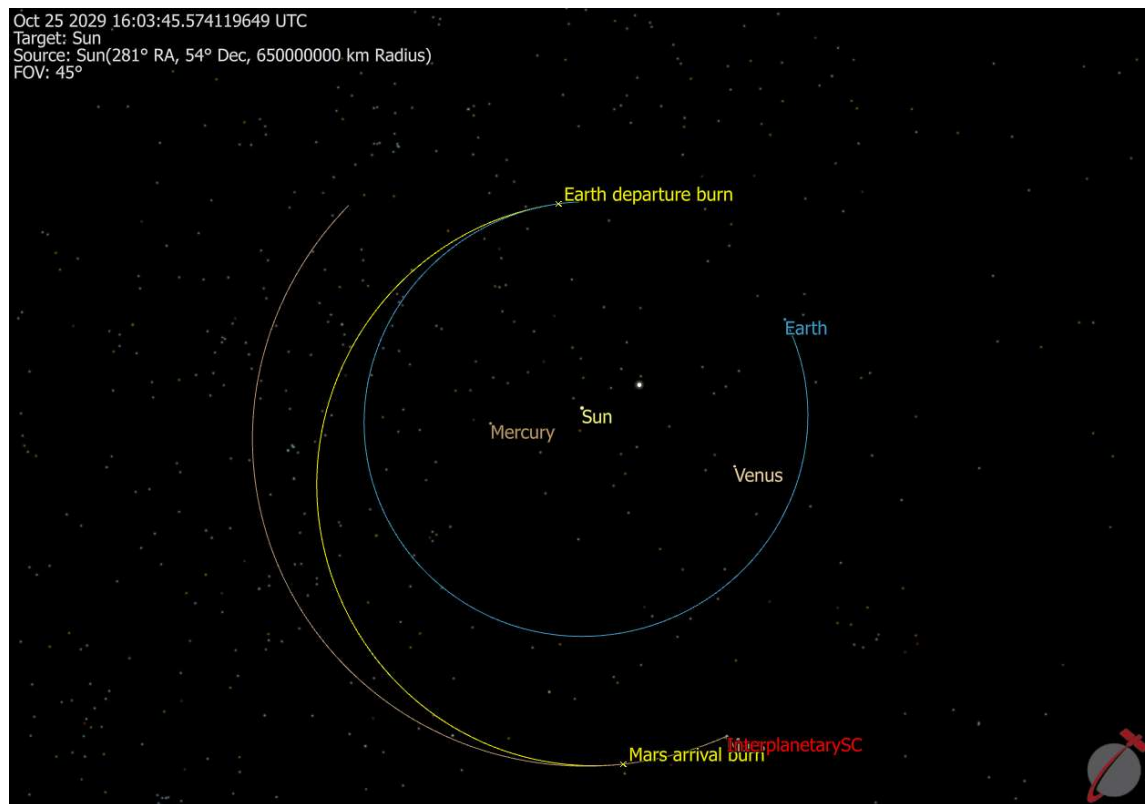


Figure 7. Simplified Hohmann transfer from Earth to Mars.

The return methods are critical to the mission to meet the planetary protection requirements. The most important factor is a safe collection at Earth. The first two design paths use the common desert drop method for sample returns. They would both employ parachutes and small thrusters to stabilize and slow their descents over Utah. An aircraft would rendezvous with the capsule in air and ensure its safe deliver to the research labs. Path 3 takes a more unusual approach, with a low thrust ion thruster installed in the sample capsule. The sample would be dropped in HEO and slow thrust down to LEO, where a SpaceX Dragon capsule would capture the sample and safely perform a splashdown landing with the sample. This could be performed with humans or robotic gear as seen fit. This plan require another launch but this could be performed at a later date without a set launch window.

Each design path selected includes a separate launch system vehicle. Design path 1 uses an Angara A5 rocket built by the Russian. The current geopolitical environment would have to be considered in this mission design. It should be noted, however, that the Angara A5 was chosen due to its smaller size and could be replaced by a larger launch system that would be unnecessarily powerful. Based on similar sized rocket launch costs, the launch cost for the Angara A5 is approximated at \$500 million. Design path 2 uses the Falcon Heavy, estimated at

\$1.6 billion for the launch cost. Path 3 uses the SLS system, which is estimated at \$3.5 billion for the launch cost. These costs estimates were based on analogous mission launch costs.

The communications system for path 1 will be a direct communication link back to Earth. The eclipse of Mars should be a short time period and the high altitude approach (over 200 km above the surface) will keep the spacecraft out of significant danger. This process will have to be automated, which is in the line of similar sample return missions like OSIRIS-Rex designed by Lockheed Martin. The lander missions, on the other hand, would use the orbital relays to communicate back with Earth. This is due to the longer expected stay for the Mars lander missions. The rover for design path 3 would also be left behind as a scientific explorer – an additional benefit of that design path – which would have to rely on the orbital relays like the other Mars rovers.

The power systems are hard to estimate due to the variety of different states of each mission path. For example, all the design paths will endure a low power travel mode between Earth and Mars. Path 1 will see a power spike when passing through the atmosphere of Mars but stay rather similar at a lower power draw. Paths 2 and 3 will see a higher power draw when landing and collecting samples due to their more active sampling methods. Path 3 also includes a rover that significantly changes the power draw if it is or is not included in the overall power budget of the system. For clarity, since that power must be considered, it has been included, though it would be a separate system when the power is being drawn by the rover.

Sub-System	Payload	Structure	Power	Comms	Propulsion	Total
Path 1 Mass (kg)	75	170	50	60	45	400
Path 1 Power (W)	30	0	80	100	10	220
Path 1 Cost (\$M)	75	30	20	25	15	165
Path 2 Mass (kg)	700	2100	450	170	1200	4620
Path 2 Power (W)	75	0	95	160	30	360
Path 2 Cost (\$M)	200	150	100	75	225	750
Path 3 Mass (kg)	1525	2050	750	195	1410	5930
Path 3 Power (W)	250	0	100	90	15	455
Path 3 Cost (\$M)	2700	200	250	100	275	3525

Table 6. Cost estimation sub-system breakdown.

The table above provides a clear sub-system breakdown of the mass and power budgets for each design path, along with their cost estimations. These are idealized numbers, with increases of about 25% for the mass and power budgets and 100% for the cost reserves and operational costs. Note the high costs for the spacecraft, which is not combined with the launch cost and operational cost of the missions. These numbers would be in the range of billions and tens of billions after they are included. While these are high budgets, this an entirely novel mission, which will naturally inflate the cost of the mission. Similar mission designs' budgets, such as NASA's current sample return mission's budget up to around \$11 billion now, must be considered against these budget estimates above. This comparison shows the budgets are reasonable for a mission of this type.

MISSION OPTIONS EVALUATION

First, the design paths detailed above must be reevaluated to confirm that they meet the mission statement, objectives, and constraints laid out at the beginning. From the table below, the design paths meet the mission objectives and fit within the constraints.

	Path 1	Path 2	Path 3
Base Design	Mars aerogel flyby	Immobile Mars lander	Mars rover
Collect Mars sample with dust	Yes	Yes	Yes
Launch in 2028/2029 window	Yes	Yes	Yes
Single launch vehicle	Yes	Yes	Yes (not including collection rendezvous)
Mars Planetary Protection protocols met	Yes	Yes	Yes

Table 7. Verifying missions meet the original mission statement, objectives, and constraints.

To select the optimal mission architecture, the design paths are evaluated against a set of weighted criteria derived from the mission objectives and constraints. The criteria are weighted to prioritize the most critical aspects of mission of success. The design paths are given a score from 1 (worst) to 5 (best) for each criteria. Then, the scores are multiplied by the criteria weight and summed to yield the weighted score for each design path option.

Design Path	1	2	3
Mission Success Likelihood (4)	4	3	2
Total Mission Cost (3)	4	2	1
Technical Risk & Maturity (2)	4	3	2
Schedule & Launch Complexity (1)	3	4	4
Weighted Score	39	28	19

Table 8. Weighted Pugh matrix to determine optimal mission architecture.

From the weighted Pugh matrix above, the aerogel flyby mission is the best choice. This is due to the higher likelihood of mission success, which is impacted by the significantly fewer points of failure for the mission design due to its simpler design. Further, the mission design is much cheaper than the alternatives, making it easier to pitch to the government and taxpayers.

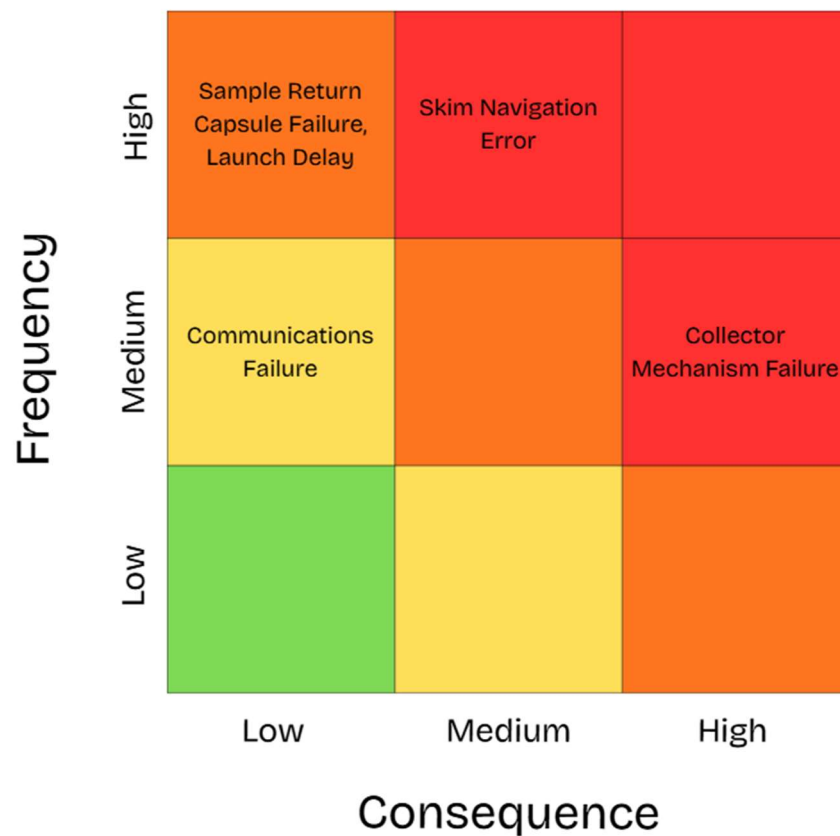


Figure 8. Risk matrix for design path.

The risks in the above risk matrix are a mix of general interplanetary risks and risks particular to this mission design. The top risks include launch delays. The free return trajectory is extremely sensitive to delays and would require a tight launch window for proper execution. This is especially critical because weather events could be entirely unavoidable. Similarly, a primary burn error would jeopardize the entire mission, since it is designed with a single, primary burn. This primary burn error is greatest risk for the mission. The sample return capsule could fail upon reentry, burning up the sample and violating the planetary protection protocols. A communications failure could leave the mission without human-in-the-loop control, which is not as great of an issue as the other design paths due to this mission design relying less on critical control moments. Finally, there could be a collector mechanism failure on deployment or sealing so the sample is not collected and secured. These are the general risks. The more unique risk to the flyby mission is a Mars atmospheric skim navigation error, where an altitude above the target would not collect the sample and a lower altitude than the target could cause the spacecraft to crash into Mars.

Risk
Primary Burn Error
Atmospheric Skim Navigation Error
Sample Return Capsule Failure
Collector Mechanism Failure
Launch Delay

Table 9. Top mission risks.

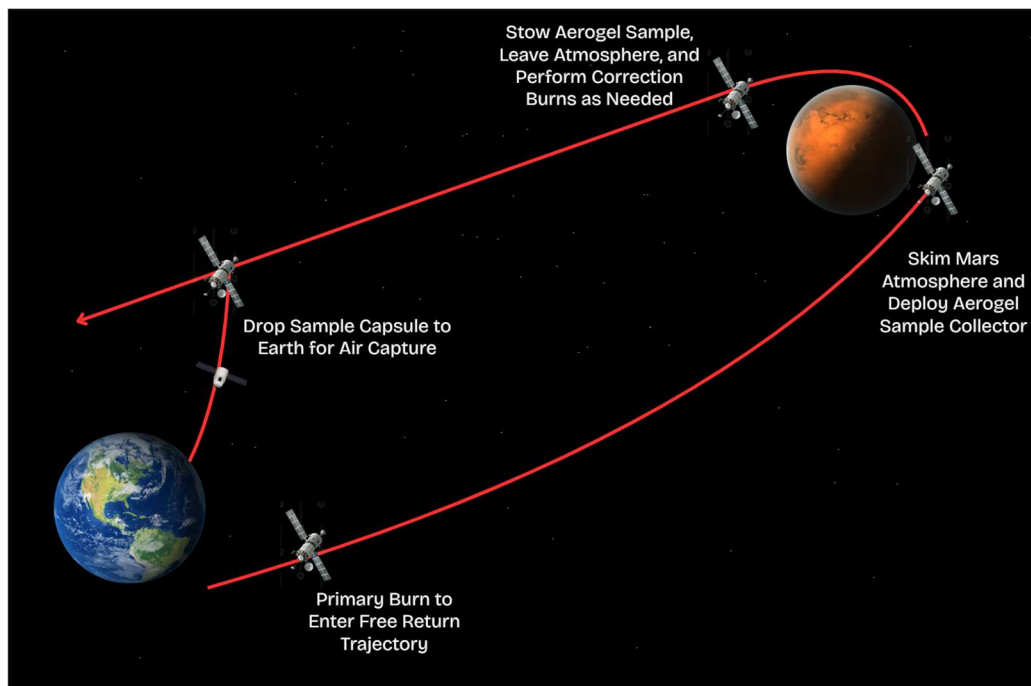


Figure 9. Mission concept of operations for flyby.

The mission conops above shows the four major steps in the mission sequence. First is the primary burn that drives the entire mission trajectory, to Mars and back. Then, at Mars, the spacecraft enters the atmosphere and deploys the aerogel collector. After leaving the atmosphere, the aerogel collector is stowed and correction burns are performed as needed. They can be performed throughout, but the drag through the atmosphere is the point with the greatest chance of drift from the designed trajectory. Finally, the sample capsule is dropped over the Utah desert for an air rendezvous to be taken safely back to a lab for further study.

DISCUSSION OF MISSION DRIVERS

Mission drivers are key parameters that have a disproportionate effect on the system's architecture, performance, and cost. A successful mission design relies on understanding and balancing mission drivers. These drivers are not independent variables but rather pieces of a tightly coupled system where a decision in one place causes a chain reaction across constraints and limitations elsewhere. The following table identifies these core drivers and their inherent relationships.

Driver	What Limits Driver	What Driver Limits
Size	Bay size, available weight, aerodynamic drag	Payload size
Launch Mass	Trajectories, Mars orbit, launch vehicle	Sample mass, payload mass, design and manufacturing cost
Power	Size, weight	Payload design, system sensitivity, mission life
Communications	Mars orbit, power	Mars orbit, timeliness, ability to command
Navigation	Cost, weight	Overall system accuracy
Scheduling	Timeline and operations, communications	Trajectory, mission utility, launch system
Operations	Cost, crew size, communications	Principal cost driver, error source

Table 10. Identification of mission drivers.

The primary mission objective, returning an atmospheric dust sample, is constrained by the requirement for a single launch architecture. This constraint makes the total launch mass and size the most dominant drivers. These drivers in turn force a tradeoff between scientific return, heavily correlated with sample size and selection, and mission complexity. The large scientific return missions are also high mass missions that require a surface landing. This greatly drives up mass and increases points of failure to the mission. Instead, a light weight, simple mission architecture, as shown by the flyby mission selected, prevails. This architecture minimizes launch mass and points of failure, in turn creating the most cost effective mission design for a sample return mission. The flyby mission has navigation as a dominant driver due to its largest point of failure, the primary burn. As stated earlier, this is the greatest risk to the mission. Ultimately, the Mars sample return mission design requires navigating the tradeoffs and chain reactions of these mission drivers. For this mission, mass and risk become the deciding factors

due to the constraint of the single launch vehicle. This ends up with the flyby aerogel mission architecture. By minimizing mass and risk, it trades the scientific return with mission success likelihood and financial viability. As seen in the analysis above, this best fulfills the primary mission objective within the given constraints.

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